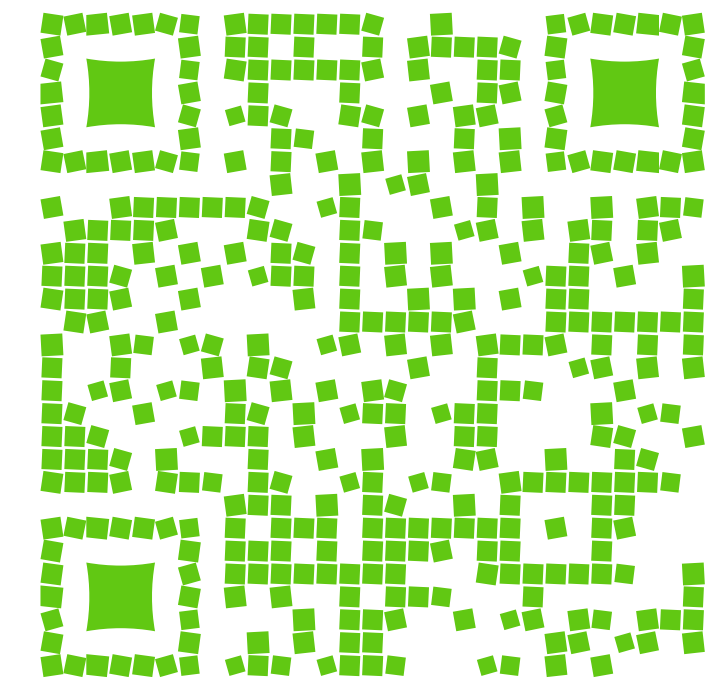


bscm: Bayesian synthetic control models in R

Causal inference with panel data

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Introduction

Panel data of $J + 1$ units observed over T time points, with unit $J + 1$ receiving treatment after T_0 . The target estimand is the causal effect

$$\tau_t = Y_{J+1,t}(1) - Y_{J+1,t}(0), \quad t > T_0.$$

Problem: Potential outcome $Y_{J+1,t}(0)$ is unobserved for $t > T_0$.

Standard synthetic control method (SCM)

SCM estimates the untreated counterfactual as a convex combination of untreated donor outcomes:

$$\hat{Y}_{J+1,t}(0) = \sum_{j=1}^J w_j y_{j,t}, \quad w \in \Delta^{J-1}$$

where w minimizes pre-treatment mean squared error.

Drawback: SCM lacks intrinsic uncertainty quantification.

Bayesian synthetic control model (BSCM)

BSCM models the untreated potential outcomes as

$$Y_{J+1,t}(0) = \alpha + x'_{J+1,t}\beta + w'_{J+1,t}\gamma_t + z'_t\omega + u_t, \quad u_t \sim N(0, \sigma^2),$$

where $\omega \in \Delta^{J-1}$, x, w are covariates, and $z_{j,t} = y_{j,t} - x'_{j,t}\beta - w'_{j,t}\gamma_t$ are covariate-adjusted donor outcomes.

The model is motivated by a low-dimensional factor representation of untreated potential outcomes and the assumption that the treated unit's factor loadings lie approximately in the convex hull of the donors' loadings.

Bayesian uncertainty quantification

Draws from the posterior predictive distribution

$$\tilde{Y}_{J+1,t}(0) \sim p(\tilde{Y}_{J+1,t}(0) \mid \text{data})$$

yield posterior draws of the treatment effect

$$\tilde{\tau}_t = y_{J+1,t} - \tilde{Y}_{J+1,t}(0)$$

$\Rightarrow$ Probabilistic treatment-effect uncertainty.

The bscm package

- Time-varying covariates, multiple treated units, AR(1) errors
- Time-varying coefficients by Bayesian P-splines
- Posterior sampling with Stan using `rstan`
- Customizable priors with weakly informative defaults
- Leave-one-out and leave-future-out cross-validation
- Placebo treatments and projection predictive inference
- On CRAN and GitHub: github.com/helske/bscm

```
fit <- bscm(
  outcome ~ x + tv(~ z), data = d,
  treatment = "treated", time = "year", unit = "id",
  priors = list(omega = logistic_normal_pr(2))
)
treatment_effect(fit)
```

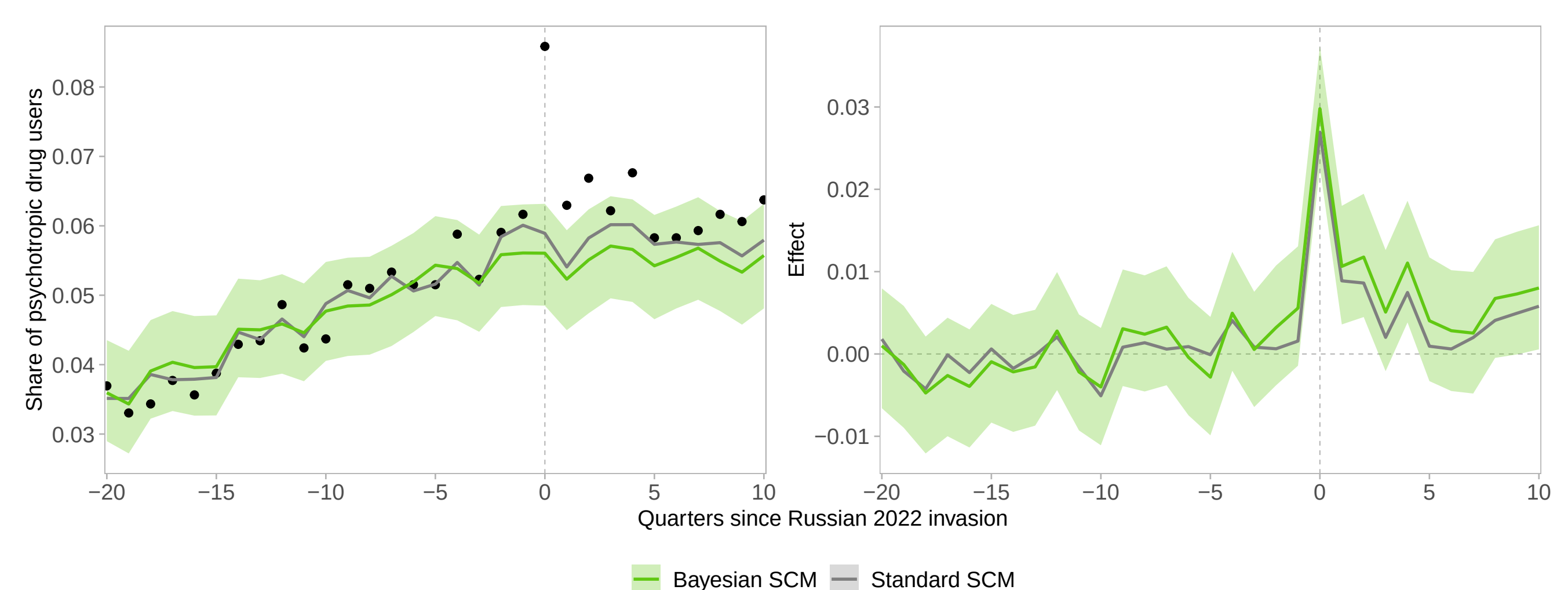
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Indirect effects of Russian 2022 invasion of Ukraine

with Thang Dang & Henri Salokangas

Did the invasion affect psychotropic medication use among Ukrainian immigrants in Finland?

- Full-population register data on immigrants entering Finland in 1983–2016.
- Medication records aggregated into 90-day periods: $T_0 = 20$.
- Ukraine + 41 donor countries; adjacent countries excluded.



Synthetic psychotropic medication use among Ukrainian immigrants (left) and estimated effects of the 2022 Russian invasion (right). Black points are observed outcomes.

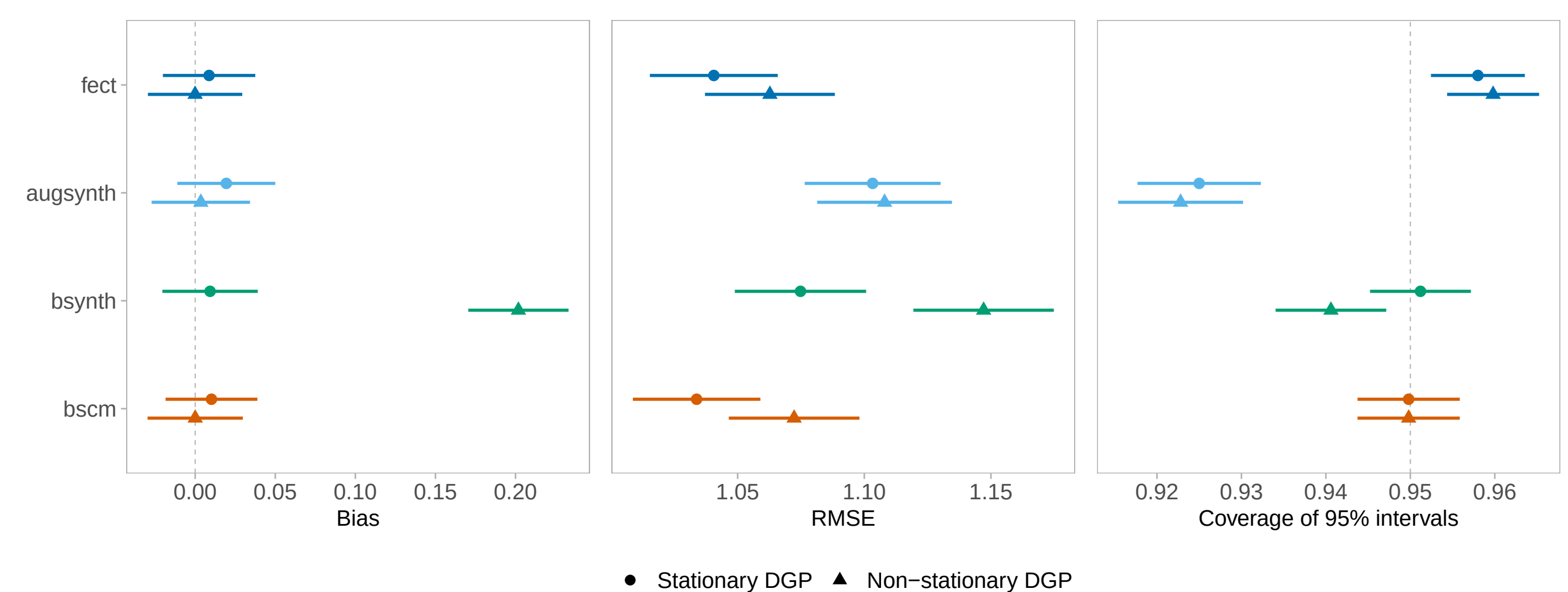
Russian invasion caused an immediate 3 percentage-point increase in psychotropic medication use among Ukrainian immigrants in Finland (95% posterior interval: 2.2–3.7).

Simulation experiments

Data generating process:

$$y_{i,t} = \alpha_i + \delta_t + \sum_{k=1}^K \lambda_{i,k} \psi_{k,t} + \tau_t \mathbf{I}(i = 1 \wedge t > T_0) + \varepsilon_{i,t}$$

with $K = 3$, $T_0 = 80$ and $J = 50$. Target is $\tau_{T_0+1} = 1$.



Bias, RMSE, and coverage probability for τ_{T_0+1} over 5000 replications, with 95% Monte Carlo error intervals. Stationary DGP: AR(1) factors with centered loadings. Nonstationary DGP: nonstationary factors with nonnegative loadings.

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